COMPUTER NETWORKS PROJECT

EXTERNAL AUTHENTICATION USING RADIUS

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Aim

To install, configure and test a FreeRADIUS server and client in Ubuntu Linux and study secure user authentication using password hashing techniques.

Introduction

Remote Authentication Dial-In User Service (RADIUS) is a networking protocol used for centralized Authentication, Authorization and Accounting (AAA). It is widely used in enterprise networks, wireless networks, VPNs and Internet Service Providers for secure user authentication.

FreeRADIUS is an open-source implementation of the RADIUS protocol. It authenticates users by verifying their credentials stored in the server database. Passwords can also be stored in hashed form for improved security.

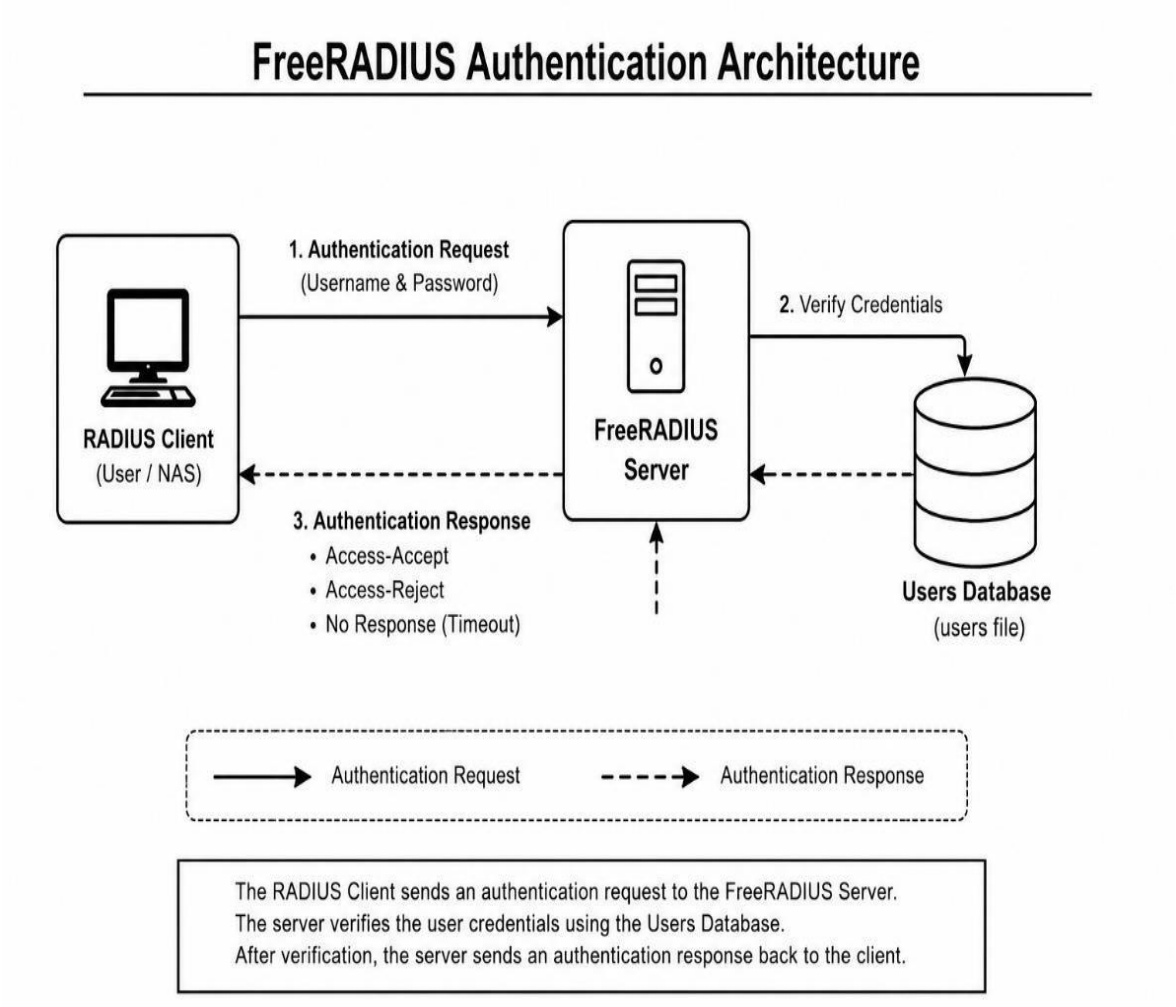
Objectives

- To install FreeRADIUS server.
- To configure FreeRADIUS server.
- To configure RADIUS client.
- To authenticate users.
- To study password hashing methods.
- To perform authentication test cases.

Software Requirements

- UBUNTU 26.04
- Oracle VirtualBox

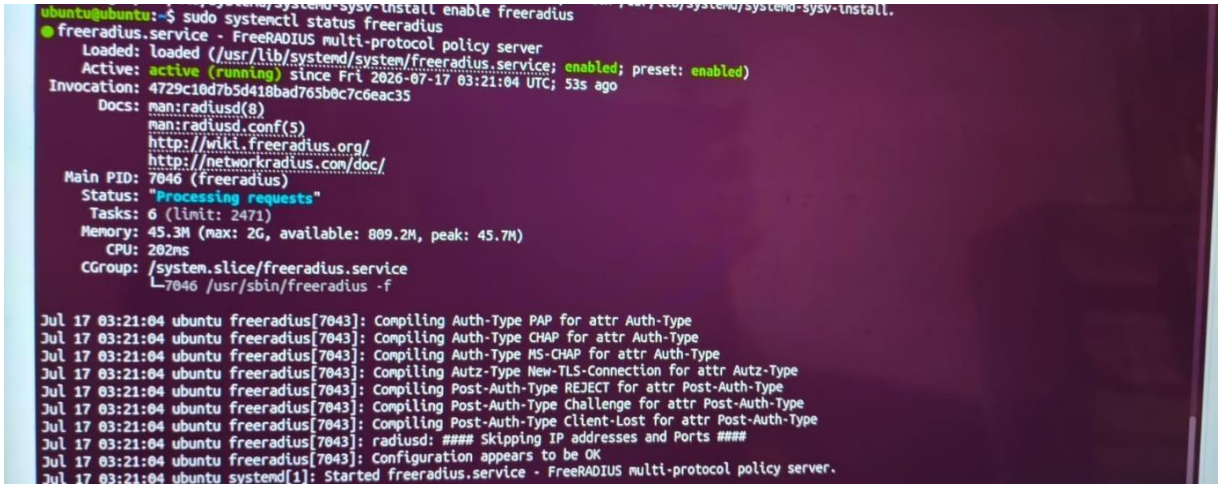
System Architecture



The above architecture shows the interaction between the user, RADIUS client, and FreeRADIUS server during the authentication process.

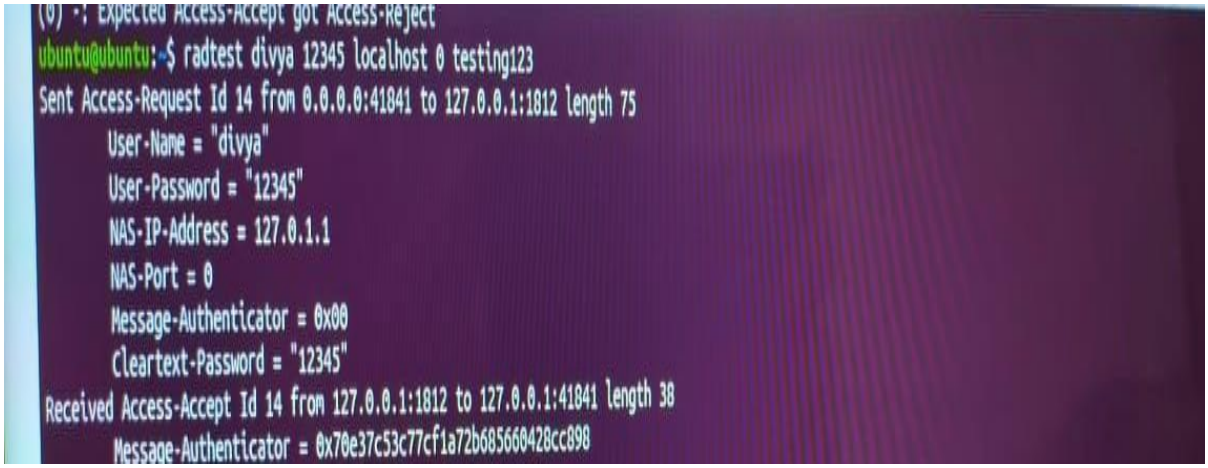
1. COMMANDS TO RUN RADIUS SERVER

The following command is used verify the status of the FreeRADIUS server



2. COMMANDS TO RUN RADIUS CLIENT

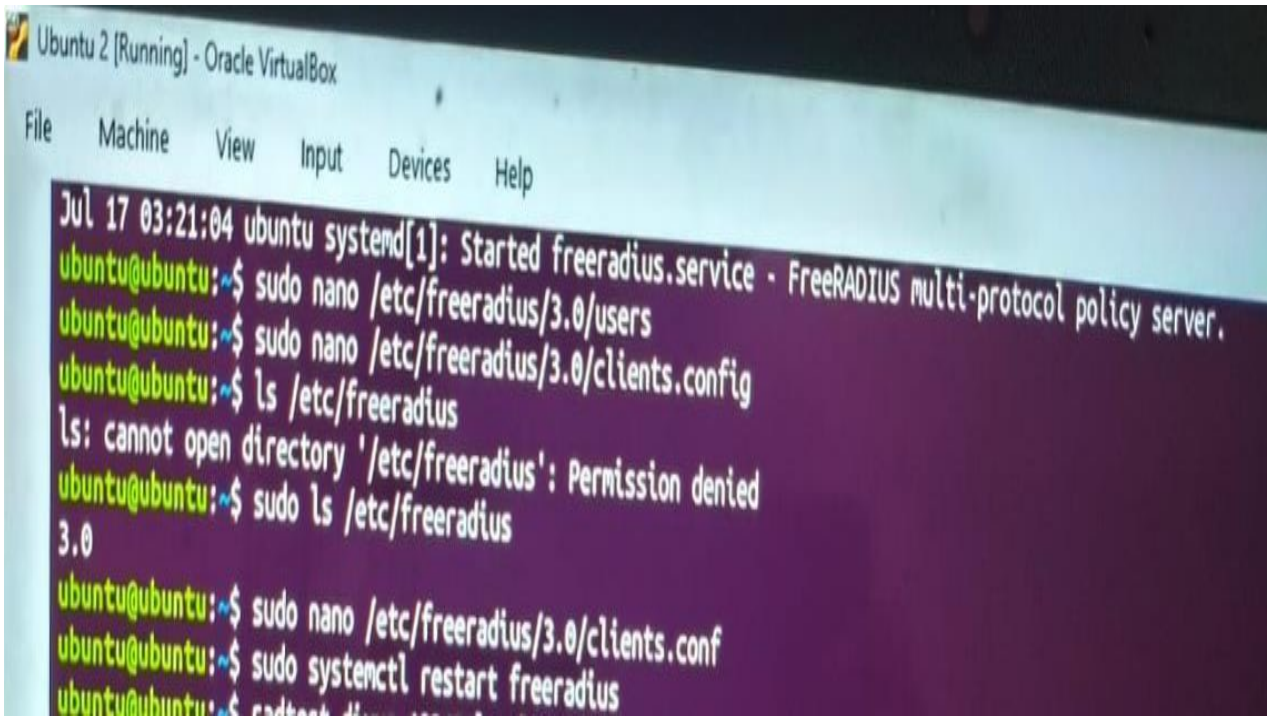
The screenshot shows the execution of the radtest command. The client successfully communicates with the FreeRADIUS server and receives the authentication response.

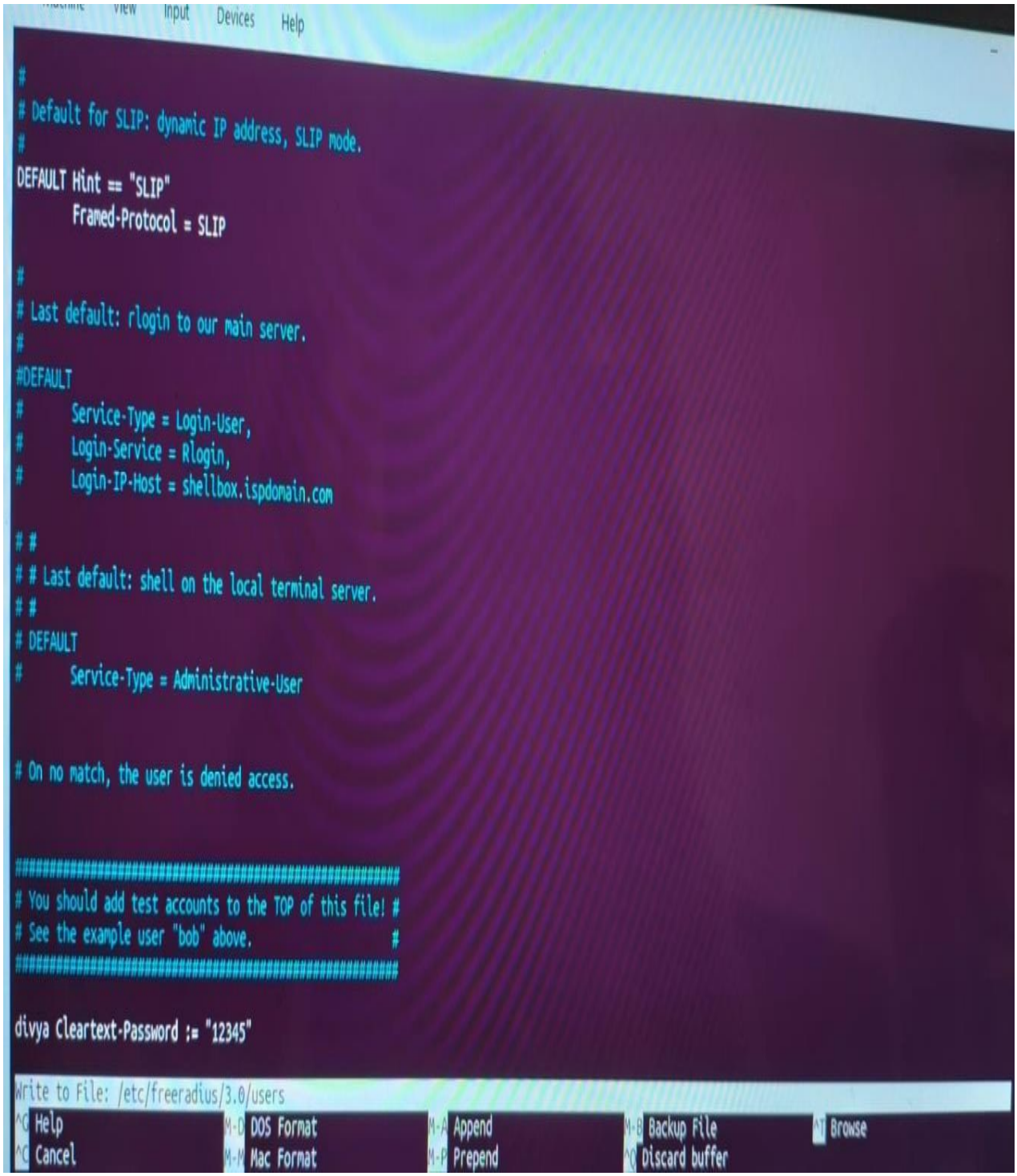


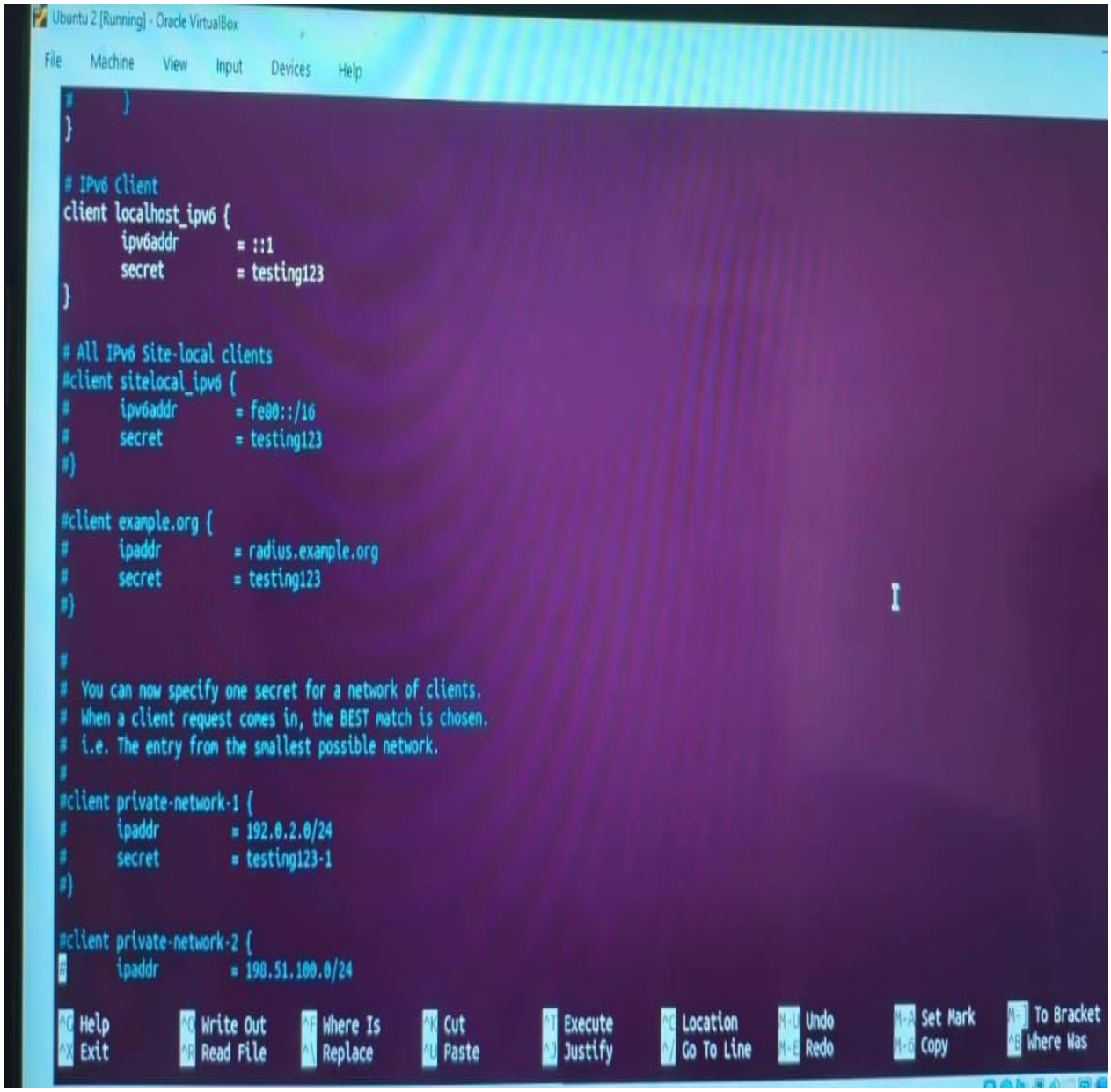
3.CONFIGURE FILE RADIUS SERVER

The clients.conf file contains the IP address and shared secret of authorized RADIUS clients. It defines which clients are permitted to communicate with the FreeRADIUS server. During authentication, the server verifies the client's IP address and shared secret before processing any authentication requests. This ensures that only trusted clients can access the RADIUS server.

The users file stores user credentials and authentication rules used by the FreeRADIUS server. It contains the usernames, passwords, and authorization attributes required for user authentication. During the authentication process, the server matches the credentials provided by the client with the entries in this file to determine whether to grant or deny access.




View Input Devices Help
#
Default for SLIP: dynamic IP address, SLIP mode.
#
DEFAULT Hint == "SLIP"
Framed-Protocol = SLIP
#
Last default: rlogin to our main server.
#
#DEFAULT
Service-Type = Login-User,
Login-Service = Rlogin,
Login-IP-Host = shellbox.ispdomain.com
# #
Last default: shell on the local terminal server.
# #
DEFAULT
Service-Type = Administrative-User
On no match, the user is denied access.
##
You should add test accounts to the TOP of this file!
See the example user "bob" above.
##
divya Cleartext-Password := "12345"
Write to File: /etc/freeradius/3.0/users
^G Help
^C Cancel
M-D DOS Format
M-M Mac Format
M-A Append
M-P Prepend
M-B Backup File
^Q Discard buffer
^T Browse



4.TEST CASES

Test Case 1 – Valid Credentials

The authentication request is sent with valid credentials. The FreeRADIUS server successfully verifies the user and returns an Access-Accept response.

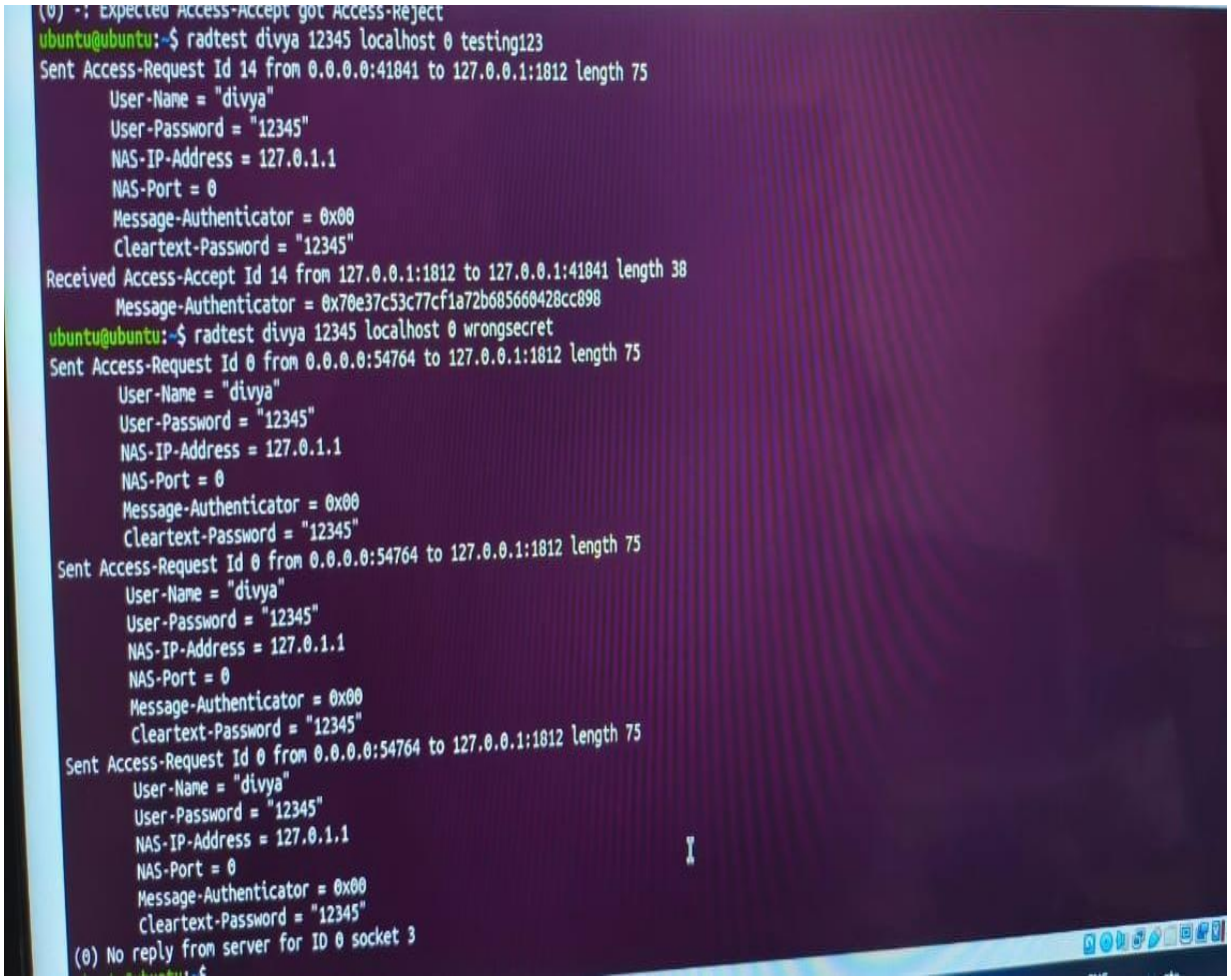
Test Case 2 – Invalid Credentials

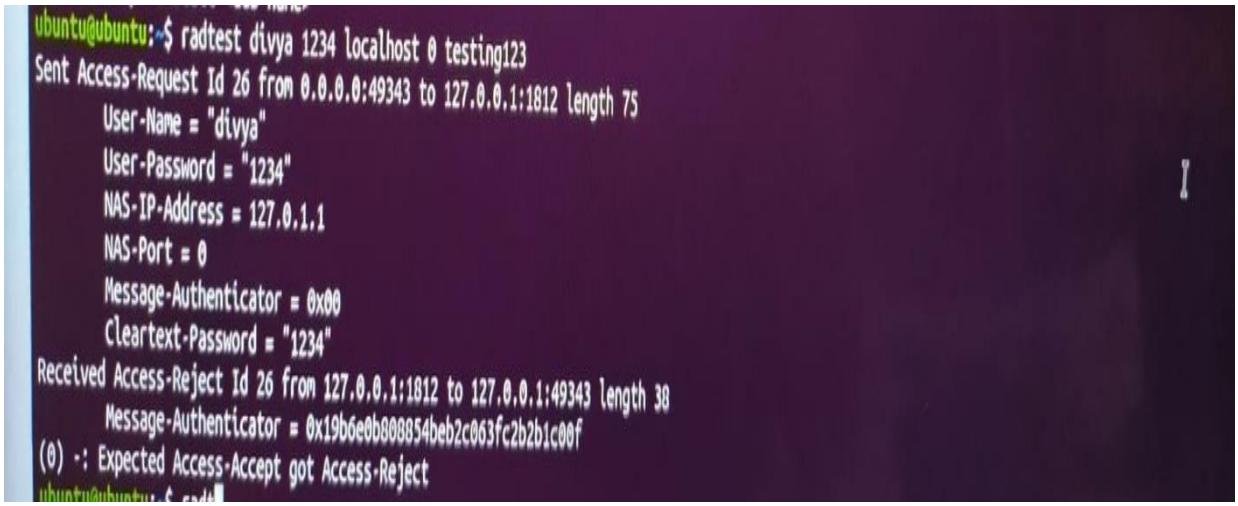
The authentication request is sent with incorrect credentials. The FreeRADIUS server rejects the request and returns an Access-Reject response.

Test Case 3 – No Reply

If the RADIUS server is unreachable, not running, or the client configuration (such as IP address or shared secret) is incorrect, the client does not receive any response.

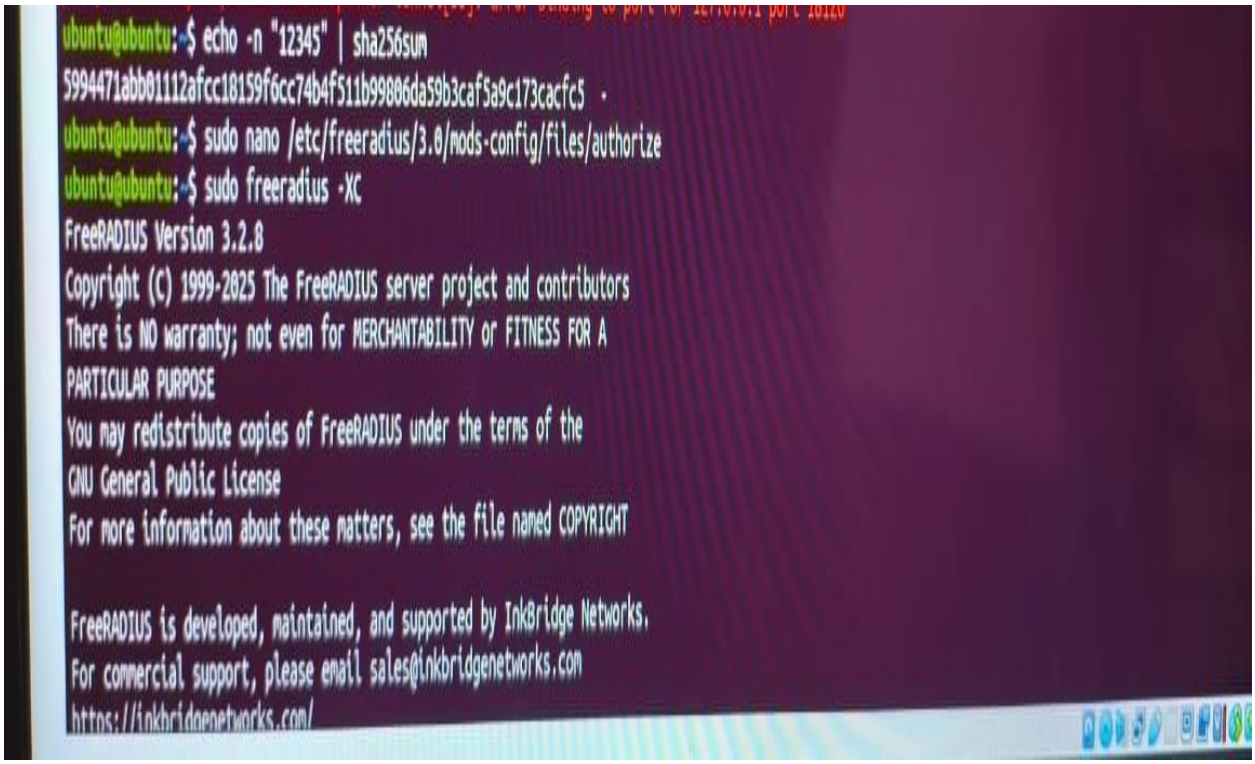
This results in a No Reply condition, indicating that the authentication request could not be processed.

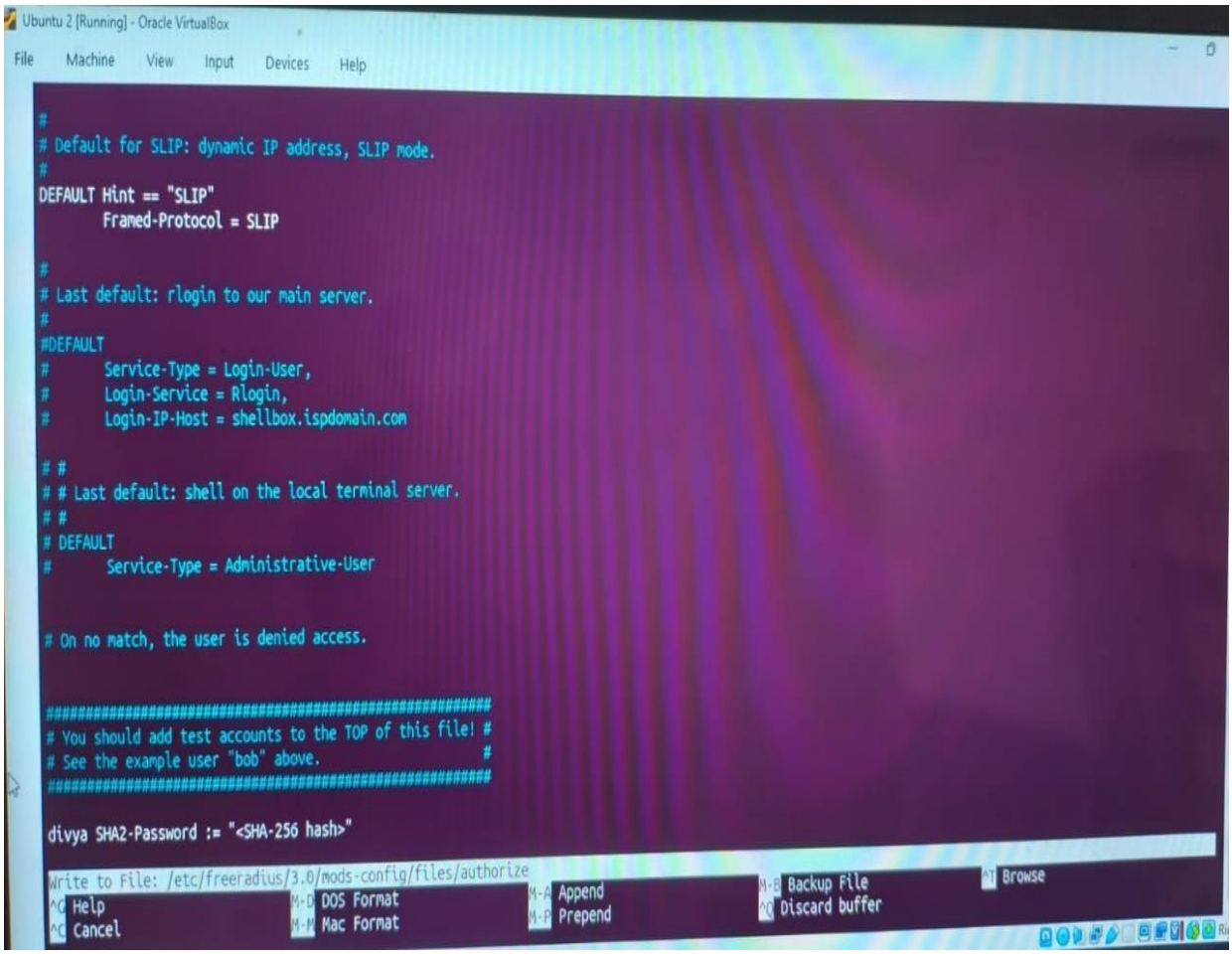




5. HASHING METHOD

The screenshot demonstrates the SHA-256 hashing process. The original password is converted into a fixed-length hexadecimal hash, which is stored securely. During login, the entered password is hashed again and matched with the stored hash.





Conclusion

The FreeRADIUS server was successfully installed, configured, and tested in the Ubuntu Linux environment. The RADIUS client was able to communicate with the server, and user authentication was verified using the **radtest** command. Password hashing methods such as SHA-256 was also studied to understand secure password storage and verification. This experiment helped in understanding the working of the RADIUS protocol and the practical implementation of centralized authentication using FreeRADIUS.